

DBMS Project Synopsis

1. Title Page

Project Title:

Crime Pattern Analysis & Smart Patrol Planning System

Course Name & Code:

UCS310 – Database Management Systems

Degree & Year:

B.Tech (2nd Year)

Department / Institute Name:

Computer Science Engineering

Lab Instructor Name:

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2. Introduction

With the rapid growth of cities and increasing population, managing public safety has become more challenging. A large amount of crime data is generated daily, but in many cases it is not used effectively for analysis and decision making. Most crime records are stored in scattered files or basic systems that do not allow proper data analysis.

This project focuses on developing a database-driven system that stores and analyses crime records to identify patterns such as crime hotspots, peak crime timings, and common crime types. The motivation behind selecting this topic is to demonstrate how a Database Management System can transform raw data into meaningful insights for better planning.

Using a DBMS instead of a file-based system ensures:

- Data consistency and integrity
- Faster data retrieval
- Reduced redundancy
- Secure and centralized data storage

3. Problem Statement

Crime data is often maintained in isolated records and spreadsheets, which makes analysis difficult and time-consuming. Due to the absence of a centralized database:

- Crime hotspots are difficult to identify
- Historical data is not used effectively
- Patrol planning becomes inefficient
- Data redundancy and inconsistency may occur

This project aims to design a centralized database system that stores crime records and helps analyse them for better decision making and planning.

4. Objectives of the Project

- To design a database using ER data modelling
- To convert the ER model into relational tables
- To apply normalization up to 3NF/BCNF
- To implement the database using SQL
- To use PL/SQL constructs such as procedures, triggers, functions and cursors

- To ensure data consistency using constraints and transactions

5. Scope of the Project

The project focuses on the backend database system and includes the following users:

Users:

- **Admin:** Responsible for managing police stations, crime categories, patrol units, and system configuration.
- **Police Staff:** Responsible for recording crime incidents, managing patrol assignments, and accessing analytical reports.

Modules Covered:

- Crime record management
- Police station management
- Patrol assignment
- Crime analytics and reporting

The project focuses on backend database design, implementation, and analytical reporting. Frontend visualization and real-time surveillance features are excluded from the project scope.

6. Proposed System Description

The proposed system will act as a centralized database for managing crime-related data. Authorized users will be able to add, update, and analyse crime records.

Key Features:

- Storage of crime details (type, location, date, status)
- Management of police station and patrol information
- Generation of reports and analytical insights

- Identification of crime hotspots and trends

The system improves:

- Efficiency through automated record management
- Consistency through constraints and normalization
- Data integrity through triggers and transactions

7. Database Design

The ER model represents the main entities, their attributes, relationships and constraints of the system.

Entity–Relationship (ER) Diagram

1. Entities & Attributes

- Crime - CrimeID (PK), CrimeType, Date, Time, Severity, Status, LocationID (FK), StationID (FK)
- Location - LocationID (PK), AreaName, Zone, City
- PoliceStation - StationID (PK), StationName, Address, ContactNumber • Officer - OfficerID (PK), OfficerName, Rank, PhoneNumber, StationID (FK) • PatrolAssignment - PatrolID (PK), PatrolDate, ShiftTime, AreaCovered, OfficerID (FK)

2. Relationships & Cardinality

- Location - Crime: One location can have many crimes; each crime occurs at one location (1:N)
- PoliceStation - Crime: One station handles many crimes; each crime belongs to one station (1:N)
- PoliceStation - Officer: One station employs many officers; each officer belongs to one

station (1:N)

- Officer - PatrolAssignment: One officer can have many patrol assignments; each assignment is given to one officer (1:N)

3. Constraints

- Primary and Foreign Keys ensure entity and referential integrity
- Important fields are NOT NULL
- Severity values restricted to Low /Medium / High

Relational Schema

Conversion of ER Diagram into Tables with foreign keys and primary keys

The following tables are derived from the ER diagram:

1. Crime(CrimeID, CrimeType, Date, Time, Severity, Status, LocationID, StationID)

- **Primary Key:** CrimeID
- **Foreign Keys:**

LocationID → references, Location(LocationID)

StationID → references, PoliceStation(StationID)

2. Location(LocationID, AreaName, Zone, City)

- **Primary Key:** LocationID

3. PoliceStation (StationID, StationName, Address, ContactNumber)

- **Primary Key:** StationID

4. Officer (OfficerID, OfficerName, Rank, Phone Number, StationID)

- **Primary Key:** OfficerID
- **Foreign Key:** StationID → references PoliceStation(StationID)

PatrolAssignment (PatrolID, PatrolDate, ShiftTime, AreaCovered, OfficerID)

- **Primary Key:** PatrolID
- **Foreign Key:** OfficerID → references, Officer (OfficerID)

Table Relationships

- Each Crime is linked to one Location and one PoliceStation.
- Each PoliceStation can have multiple Officers.
- Each Officer can be assigned multiple PatrolAssignments.

All foreign key relationships enforce referential integrity, ensuring that related records exist before insertion or update.

8. Normalization

Functional dependencies will be identified and tables will be normalized:

- 1NF: Removal of repeating groups
- 2NF: Removal of partial dependencies
- 3NF: Removal of transitive dependencies

1.Crime

Field Name	Data Type	Constraints
CrimeID	INT	Primary Key
CrimeType	VARCHAR(50)	NOT NULL
Date	DATE	NOT NULL
Time	TIME	NOT NULL
Severity	VARCHAR(10)	CHECK (Low/Medium/High)
Status	VARCHAR(20)	NOT NULL
LocationID	INT	Foreign Key

StationID INT Foreign Key

2. Location

Field Name	Data Type	Constraints
LocationID	INT	Primary Key

AreaName	VARCHAR(50)	NOT NULL
Zone	VARCHAR(50)	NOT NULL
City	VARCHAR(50)	NOT NULL

3. Police Station

Field Name	Data Type	Constraints
StationID	INT	Primary Key
StationName	VARCHAR(50)	NOT NULL
Address	VARCHAR(100)	NOT NULL
ContactNumber	VARCHAR(15)	NOT NULL

4. Officer

Field Name	Data Type	Constraints
OfficerID	INT	Primary Key
OfficerName	VARCHAR(50)	NOT NULL
Rank	VARCHAR(30)	NOT NULL
PhoneNumber	VARCHAR(15)	NOT NULL
StationID	INT	Foreign Key

5. Patrol Assignment

Field Name	Data Type	Constraints
PatrolID	INT	Primary Key
PatrolDate	DATE	NOT NULL
ShiftTime	VARCHAR(20)	NOT NULL
AreaCovered	VARCHAR(50)	NOT NULL

OfficerID	INT	Foreign Key
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9. Database Implementation

9.1 SQL Implementation

SQL will be used to create and manage the database and perform analytical queries.

- **DDL:** Create tables and apply primary key, foreign key, NOT NULL and CHECK constraints.
- **DML:** Insert, update and delete crime, officer and patrol records.
- **Queries:** Use joins, subqueries and aggregate functions to identify crime hotspots, peak crime hours and crime trends.
- **Views:** Create views for summarized crime reports and analytics.

9.2 PL/SQL Components

The project will implement:

1. Stored Procedures

Stored procedures will be used to automate repetitive database operations and report generation.

Examples:

- Procedure to generate crime summary reports by area and time
- Procedure to assign patrol teams to high-crime areas

2. Functions

User-defined functions will be created to perform calculations and return useful statistics.

Examples:

- Function to calculate total crimes in a specific area

- Function to compute risk level based on crime frequency and severity

3. Triggers

Triggers will be implemented to automatically enforce rules and maintain data integrity.

Examples:

- Trigger to validate crime severity values before insertion
- Trigger to log updates made to crime records

4. Cursors

Cursors will be used to process multiple records sequentially when generating analytical reports and summaries.

5. Exception Handling

Exception handling mechanisms will be included to manage runtime errors and ensure reliable database operations.

10. Transaction Management & Concurrency

- Transaction management will be implemented to ensure that crime and patrol data remains accurate and consistent during database operations.
- Use of COMMIT and ROLLBACK to maintain data consistency while inserting or updating crime records.
- Use of SAVEPOINTS to recover partial transactions in case of errors.
- Ensuring ACID properties (Atomicity, Consistency, Isolation, Durability) during operations such as adding crime reports and assigning patrol teams.
- Basic handling of concurrent access to prevent conflicts when multiple users update records simultaneously.

11. Tools & Technologies Used

- **MySQL:** Database creation, queries, triggers, procedures and views.
- **SQL & PL/SQL:** Data manipulation, analytical queries and automation.
- **MySQL Workbench:** Database design and query execution.
- **Python (Flask):** Backend to connect database with the interface.
- **HTML, CSS, JavaScript:** Simple web interface.

- **Chart.js:** Data visualization and charts.
- **VS Code:** Development environment.

12. Expected Outcomes

- Development of a fully **normalized relational database** for crime data management.
- **Efficient data retrieval and analysis** using SQL queries, joins and aggregate functions.
- **Automation of database operations** using triggers, stored procedures and functions.
- Improved **data consistency and integrity** through constraints and referential integrity.
- Generation of **analytical reports and views** for identifying crime trends and hotspots.
- Demonstration of the **practical application of DBMS concepts** in solving real-world problems.